25900559 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEPHY201 - COMPUTATIONAL PHYSICS - C++ PROGRAMMING

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **2** marks

1. Why are << and >> operators used in C++ programming? [K] / [CO1]
2. Define keywords and give four examples of C++ keywords. [K] / [CO1]
3. From the given list (C, C++, FORTRAN, Python, COBOL, Java), classify the languages into object-oriented and procedure-oriented categories. [K] / [CO1]
4. Write a for loop to print numbers from 10 down to 1. [U] / [CO2]
5. Write the difference between break and continue statements in C++. [K] / [CO2]
6. What is the function of the parameter list in C++? [K] / [CO2]
7. What are one dimensional arrays? [K] / [CO2]
8. Mention any two category of functions with examples [K] / [CO2]
9. What values are given inside the brackets of a function call? [K] / [CO2]
10. State the advantages of using operator overloading in C++. [K] / [CO3]

11. What do you mean by an array containing objects of a class? [K] / [CO3]

12. What are the benefits of making an array a member of a class? [U] / [CO3]

13. What do you mean by a pointer in C++ programming? [K] / [CO3]

14. Explain the concept of static member functions in a class. [K] / [CO3]

[2x10 = 20]

Part B

Short Essay Type Questions

Answer any **6** questions

Each question carries **5** marks

15. Create an algorithm that determines which of three numbers is the largest. [U] / [CO1]

16. Write a C++ program to demonstrate logical AND, OR, and NOT operations. [U] / [CO1]

17. Define a nested loop in C++ and demonstrate its working using a simple program. [K] / [CO2]

18. (a) What is the purpose of an if statement in C++ ? (b) Write the basic syntax of an if statement in C++. (c) What data type must be in the condition of an if statement? [U] / [CO2]

19. Compare switch statements and if…else-if constructs in C++ and highlight the differences. [K] / [CO2]

20. Show how to declare and initialize a 2D array in C++ with a suitable example. [U] / [CO2]

21. What are the essential elements of a function in C++? Describe each briefly. [U] / [CO2]

22. Describe the steps involved in opening, reading, writing, and closing a file in C++. [K] / [CO3]

[5x6 = 30]